

A Practical Guide to Curse of dimensionality

TOPIC -- CURSE OF DIMENSIONALITY

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} r^d$$

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Machine learning In machine learning problems that involve learning a "state-of-nature" from a finite number of data samples in a high-dimensional feature space with each feature having a range of possible values, typically an enormous amount of training data is required to ensure that there are several samples with each combination of values. In an abstract sense, as the number of features or dimensions grows, the amount of data we need to generalize accurately grows exponentially. A typical rule of thumb is that there should be at least 5 training examples for each dimension in the representation. In machine learning and insofar as predictive performance is concerned, the curse of dimensionality is used interchangeably with the peaking phenomenon, which is also known as Hughes phenomenon. This phenomenon states that with a fixed number of training samples, the average (expected) predictive power of a classifier or regressor first increases as the number of dimensions or features used is increased but beyond a certain dimensionality it starts deteriorating instead of improving steadily. Nevertheless, in the context of a simple classifier (e.g., linear discriminant analysis in the multivariate Gaussian model under the assumption of a common known covariance matrix), Zollanvari et al. showed both analytically and empirically that as long as the relative cumulative efficacy of an additional feature set (with respect to features that are already part of the classifier) is greater (or less) than the size of this additional feature set, the expected error of the classifier constructed using these additional features will be less (or greater) than the expected error of the classifier constructed without them. In other words, both the size of additional features and their (relative) cumulative discriminatory effect are important in observing a decrease or increase in the average predictive power. In metric learning, higher dimensions can sometimes allow a model to achieve better performance. After normalizing embeddings to the surface of a hypersphere, FaceNet achieves the best performance using 128 dimensions as opposed to 64, 256, or 512 dimensions in one ablation study. A loss function for unitary-invariant dissimilarity between

word embeddings was found to be minimized in high dimensions.

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Therefore, the squared distance from the origin, $r^2 = \sum_i x_i^2$ has the average value $d/3$ and variance $4d/45$. For large d , distribution of r^2/d is close to the normal distribution with the mean $1/3$ and the standard deviation $2/\sqrt{45d}$ according to the central limit theorem. Thus, when uniformly generating points in high dimensions, both the "middle" of the hypercube, and the corners are empty, and all the volume is concentrated near the surface of a sphere of "intermediate" radius $\sqrt{d/3}$. This also helps to understand the chi-squared distribution. Indeed, the (non-central) chi-squared distribution associated to a random point in the interval $[-1, 1]$ is the same as the distribution of the length-squared of a random point in the d -cube. By the law of large numbers, this distribution concentrates itself in a narrow band around d times the standard deviation squared (σ^2) of the original derivation. This illuminates the chi-squared distribution and also illustrates that most of the volume of the d -cube concentrates near the boundary of a sphere of radius $\sigma \sqrt{d}$. A further development of this phenomenon is as follows. Any fixed distribution on the real numbers induces a product distribution on points in $\mathbb{R}^d$. For any fixed n , it turns out that the difference between the minimum and the maximum distance between a random reference point Q and a list of n random data points $P_1, \dots, P_n$ become indiscernible compared to the minimum distance:

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Nearest neighbor search The effect complicates nearest neighbor search in high dimensional space. It is not possible to quickly reject candidates by using the difference in one coordinate as a lower bound for a distance based on all the dimensions. However, it has recently been observed that the mere number of dimensions does not necessarily result in difficulties, since relevant additional dimensions can also increase the contrast. In addition, for the resulting ranking it remains useful to discern close and far neighbors. Irrelevant ("noise") dimensions, however, reduce the contrast in the manner described above. In time series analysis, where the data are inherently high-dimensional, distance functions also work reliably as long as the signal-to-noise ratio is high enough.

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Sampling There is an exponential increase in volume associated with adding extra dimensions to a mathematical space. For example, $10^2 = 100$ evenly

spaced sample points suffice to sample a unit interval (try to visualize a "1-dimensional" cube, i.e. a line) with no more than $10^{-2} = 0.01$ distance between points; an equivalent sampling of a 10-dimensional unit hypercube with a lattice that has a spacing of $10^{-2} = 0.01$ between adjacent points would require $10^{20} = [(10^2)^{10}]$ sample points. In general, with a spacing distance of 10^{-n} the 10-dimensional hypercube appears to be a factor of $10^n(10-1) = [(10^n)10/(10^n)]$ "larger" than the 1-dimensional hypercube, which is the unit interval. In the above example $n = 2$: when using a sampling distance of 0.01 the 10-dimensional hypercube appears to be 10^{18} "larger" than the unit interval. This effect is a combination of the combinatorics problems above and the distance function problems explained below.

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Optimization When solving dynamic optimization problems by numerical backward induction, the objective function must be computed for each combination of values. This is a significant obstacle when the dimension of the "state variable" is large.

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$\lim_{d \rightarrow \infty} E \left(\frac{\text{dist}_{\max}(d) - \text{dist}_{\min}(d)}{\text{dist}_{\min}(d)} \right) \rightarrow 0$
 $\{\displaystyle \lim_{d \rightarrow \infty} E \left(\frac{\text{dist}_{\max}(d) - \text{dist}_{\min}(d)}{\text{dist}_{\min}(d)} \right) \rightarrow 0\}$. This is often cited as distance functions losing their usefulness (for the nearest-neighbor criterion in feature-comparison algorithms, for example) in high dimensions. However, recent research has shown this to only hold in the artificial scenario when the one-dimensional distributions $\mathbb{R}$ are independent and identically distributed. When attributes are correlated, data can become easier and provide higher distance contrast and the signal-to-noise ratio was found to play an important role, thus feature selection should be used. More recently, it has been suggested that there may be a conceptual flaw in the argument that contrast-loss creates a curse in high dimensions. Machine learning can be understood as the problem of assigning instances to their respective generative process of origin, with class labels acting as symbolic representations of individual generative processes. The curse's derivation assumes all instances are independent, identical outcomes of a single high dimensional generative process. If there is only one generative process, there would exist only one (naturally occurring) class and machine learning would be conceptually ill-defined in both high and low dimensions. Thus, the traditional argument that contrast-loss creates a curse, may be fundamentally inappropriate. In addition, it has been shown that when the generative model is

modified to accommodate multiple generative processes, contrast-loss can morph from a curse to a blessing, as it ensures that the nearest-neighbor of an instance is almost-surely its most closely related instance. From this perspective, contrast-loss makes high dimensional distances especially meaningful and not especially non-meaningful as is often argued.